Palaeomagnetism of Prince Edward Island

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Summary

This paper describes the magnetic properties of the bedrock of Prince Edward Island situated in the Gulf of St. Lawrence on the eastern coast of Canada. The intensities and direction of magnetization of 60 samples have been measured. The mean declination of these samples is almost opposite to the declination of the present Earth's field at the location and the mean inclination differs from that of the Earth's field by 66°. The pole position, calculated from these data, is located in the Yellow Sea.

1. Introduction

Approximately 80 samples were collected on Prince Edward Island which is situated between lat. 45·9° and 47·1° N and long. 62·0° and 64·4° W. They consisted of dark-red sandstones and soft red shales and give a good representation of the bedrock of the Island which is classified by geologists as being of either Carboniferous–Pennsylvanian or Permian age.

2. Sampling

The samples, previously oriented in the field, were cored into cylinders of 2·4 cm diameter and, the thickness of the samples permitting, sliced to 3 cm length for reasons given later.

Since a few soft, friable red sandstones were ruined in transport or the coring process, 60 of the 80 samples were left available for measurements. They originate from 17 sites and the number of samples from each site is listed in Table 1. The location of each site is shown in Figure 1.

3. Magnetic properties

The horizontal and vertical components of intensity of magnetization together with their directions were determined for each specimen with the Dominion Observatory astatic magnetometer which has been described elsewhere (Roy 1963a). The mean horizontal, vertical and total intensities of magnetization for each site are listed in Table 1. Also given in the same table are the directions of magnetization D of the horizontal component and the dip angle I of the direction of magnetization from the horizontal. D is measured clockwise from geographic

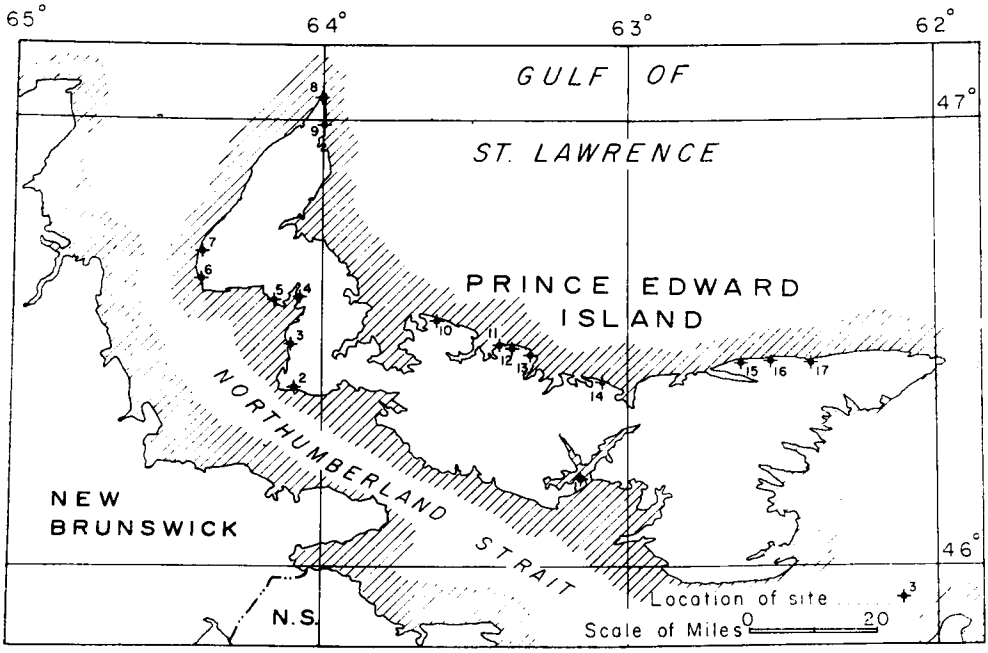


FIG. 1.—Map of Prince Edward Island showing the location of the 17 sites.

Table 1

Magnetic properties of rock specimens from Prince Edward Island. p_h, p_z and p are the horizontal, vertical and total intensities of magnetization. D is measured clockwise from geographic north and I is positive down.

Site	Specimens	p_h e.m.u./cm³ × 10⁻⁶	p_z e.m.u./cm³ × 10⁻⁶	p e.m.u./cm³ × 10⁻⁶	D deg	I deg
1	3	1·4	0·2	1·4	169	9
2	6	8·1	−0·2	8·1	167	− 1
3	4	7·8	0·7	7·8	170	4
4	2	3·9	0·2	3·9	171	3
5	3	5·1	−1·0	5·2	176	−11
6	4	4·7	0·2	4·7	165	3
7	4	4·9	0·9	5·0	172	11
8	3	3·0	1·3	3·3	203	25
9	6	1·3	0·1	1·3	206	6
10	2	2·1	0·9	2·3	198	23
11	3	1·1	0·5	1·2	160	23
12	6	2·5	0·1	2·5	160	2
13	2	2·0	−0·3	2·0	202	− 8
14	4	1·8	0·9	2·0	170	27
15	3	1·6	0·4	1·7	163	14
16	2	10·6	0·8	10·6	175	5
17	3	2·9	0·4	2·9	159	9

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north and I is positive down. As seen in Table 1, these specimens were weakly magnetized vertically and it became important to obtain the largest possible deflections in the readings so as to reduce the error in I. We have shown elsewhere (Roy 1963b) that the response of the astatic magnetometer to the vertical component of magnetization of a cylindrical rock specimen is optimum when the ratio of height to diameter is approximately equal to 1·25. Since 80 per cent of the samples were thick enough to produce cores 3 cm long, most of the individual dip determinations are within an error of 2°. The directions of magnetization of the 17 sites are plotted in Figure 2. The directions are all southerly with a very low dip and make an angle of approximately 180° with the direction of the present field. The mean direction of 8·1° N and 174·7° E is also plotted together with a circle of confidence of 7° calculated at the 95 per cent level using the statistical method of Fisher (1953). D and I represent the mean value of the direction of magnetization of each specimen rather than the directions of the mean horizontal and total intensities in order to reduce the inaccuracy of orientation of each specimen which is 4° to 5°. In the computation, the negative inclinations were not considered reversals of the field and therefore were calculated with a negative sign.

It is evident that the magnetization of these samples cannot have been produced

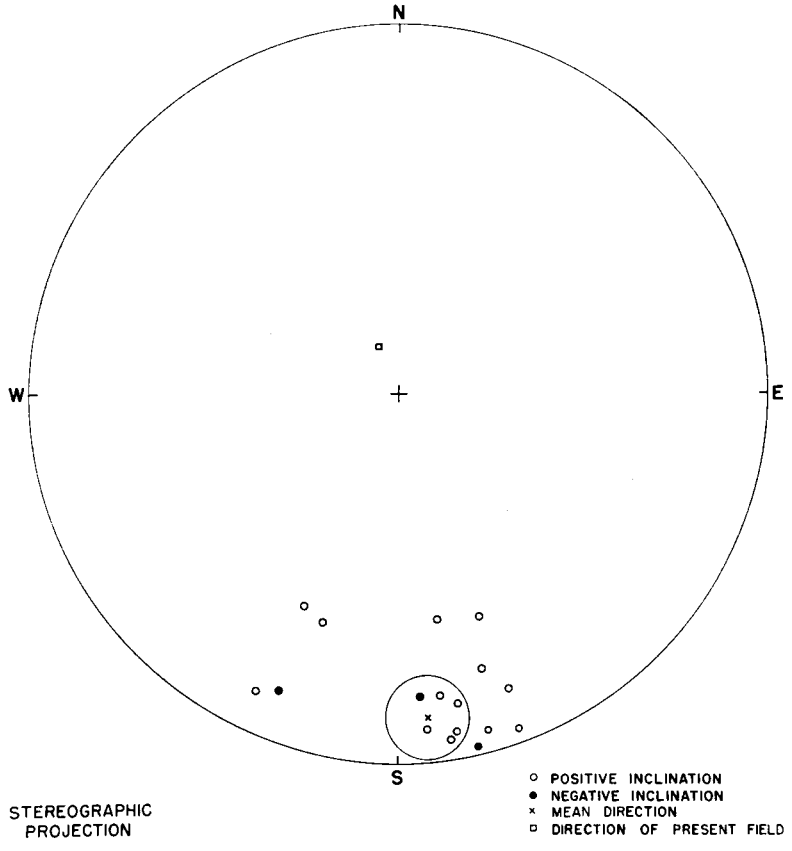


FIG. 2.—The directions of magnetization at 17 sites in Prince Edward Island.

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by the present Earth's field since the averaged magnetic directions of the field at Prince Edward Island are 74·5° N and 24·2° W as shown in Figure 2.

4. Comparison with other results

It has been the practice in palaeomagnetic studies to express the data obtained from rock specimens in terms of the geomagnetic dipole producing the field directions measured. It is assumed that the directions of the field measured over 10^4 years average out to the direction produced by the mean geomagnetic pole. From the coordinates of the site and the directions of the field, the position of the pole can then be determined. On this basis, pole positions for various geological era have been calculated by several workers from measurements made on rock specimens. Using our data, with a mean site location at 46° 30′ N and 63· °40′ W (the sites being mostly in the western half of the Island) calculations locate the north pole for that era on the north shore of the Yellow Sea at lat. 39·0° N and long. 123·2° E. The semi-axes of the 95 per cent ellipse of confidence of, and perpendicular to, the co-latitude are 3·7° and 7·4°.

From Cox and Doell (1960) who have conveniently listed the pole positions available up to 1960, we find that one of the best determined positions is obtained from the Supai Formation in western United States of America. From data given by Doell (1955), Graham (1955), Runcorn (1956) and Collinson and Runcorn (1960), Cox and Doell have calculated a mean pole position for that formation, which is classified as Permian and Pennsylvanian, at 40½° N and 110° E. From the same paper, we find two pole positions determined from locations much closer to Prince Edward Island. Dubois (1959) obtains a pole position at 30° N and 133° E from samples collected in the Gaspe Peninsula and New Brunswick while the pole position obtained from Nairn *et al.* (1959) for Newfoundland samples is located at 36·5° N and 139° E. The rocks are classified as late Mississippian and Pennsylvanian and give a mean pole position at 33° N, and 136° E.

Since our pole position lies between the pole positions obtained from these two groups, we find that the palaeomagnetic data are in close agreement with the geological dating of the bedrock of Prince Edward Island, i.e. late Pennsylvanian or possibly Permian. More magnetic measurements on known Pennsylvanian and Permian formations might make it possible to determine more accurately the age of this rock formation.

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References

Cox, A. & Doell, R. R., 1960. *Bull. Geol. Soc. Amer.*, **71,** 645.
Collinson, D. W. & Runcorn, S. K., 1960. *Bull. Geol. Soc. Amer.*, **71,** 915.
Doell, R. R., 1955. *Nature* (*Lond.*), **183,** 1183.

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Dubois, P. M., 1959. *Nature* (*Lond.*), **184,** B.A. 63.
Fisher, R. A., 1953. *Proc. Roy. Soc.* A, **217,** 295.
Graham, J. W., 1955. *J. Geophys. Res.*, **60,** 329.
Nairn, A. E. M., Frost, D. V. & Light, B. G., 1959. *Nature* (*Lond.*), **183,** 596.
Roy, J., 1963a. *Pub. Dom. Obs.*, **27,** No. 9 (in press).
Roy, J., 1963b. *Nature* (*Lond.*), **197,** 134.
Runcorn, S. K., 1956. *Bull. Geol. Soc. Amer.*, **67,** 301.

13652463b, 1963, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-246X.1963.tb06286.x by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License